PHY-275 Fall 2026 Final Project Survival Guide

Expected workload: 6 hrs

Expected outcome: Analytical, visual, computational, or experimental application of a classical mechanic topic that *you enjoyed* the most.

Here're some examples, you don't need to do control systems and robotic applications per se. It's just a low hanging fruit.

https://assets-eu.researchsquare.com/files/rs-1000853/v1_covered.pdf

https://journals.sagepub.com/doi/full/10.1177/00202940211035406

https://ctms.engin.umich.edu/CTMS/index.php?example=BallBeam&section=SystemModeling (The whole tab, from Kinematic to Control and Simulation)

https://www.researchgate.net/publication/399911073_FORWARD_KINEMATICS_FOR_THE_STATE_ESTIMATION_OF_QUADRUPED_ROBOT

Artifacts options:

- Presentation (5-7 slides, 10 min)
- GitHub repo with a detailed MD write-up
- A published GitHub site with a comprehensive readme file

Submission timeline:

Step 1: Project proposal (Week 13)

Step 2: Tech demo (or principal analysis, Week 15)

Step 3: Final presentation/Repo/Site online (Week 16)

Grading rubrics for the final project

Criteria	3 points	2 points	1 point	0 point
Physics Formulation	Governing equations, state variables, and coordinate frames are explicitly derived and accurate, adhering to standard conventions.	Core physical principles are correctly identified with minor algebraic or notation slips that do not invalidate the overall dynamics.	Major physical misconceptions; equations of motion are missing, incomplete, or decoupled from the dynamic behavior.	No physical model presented, or completely unrelated to classical mechanics.
Artifact Execution & Technical Work	Simulation, code, hardware demo, or analytical derivation executes cleanly, correctly models the physical behavior, and reflects the ~6-hour project scope.	Working implementation that produces plausible results, but features narrow parameters, rough edges, or minimal visual interpretation.	Implementation is broken, unfinished, or relies on boilerplate code without meaningful adaptation to the mechanics problem.	Deliverable does not run, lacks code/derivation, or provides no usable data or results.
Documentation & Delivery (	**Slides:** 5–7 clean, well-paced slides within the 10-minute limit.**Repo/Site:** Structured README or published page with rendered LaTeX, embedded visuals, and clear setup/run steps.	**Slides:** Covers requirements but is overly text-heavy or unevenly paced.**Repo/Site:** Functional documentation, but equations are unformatted or run instructions are vague.Repo/Site: Functional documentation, but equations are unformatted or run instructions are vague.	Slides: Disorganized, missing key components, or violates length constraints.Repo/Site: Sparse README, missing equations, or broken assets/build errors.	Deliverable is missing, broken link (e.g., 404 site error), unreadable, or empty repository.
Physical Interpretation & Limitations	Thorough discussion connecting results to mechanics concepts (e.g., energy conservation, limiting cases, stability, or damping), clearly noting assumptions and errors. Provided proper commnets and insights of the codebase.	Identifies results and basic limitations, but lacks quantitative verification against expected benchmarks or analytical limits. Proper comments of codebase.	Merely reports raw outputs with little to no interpretation of whether the behavior is physically realistic. Lacking proper comments of the codebase.	No discussion of results, error sources, or physical validity. Or cannot explain the basic function or structure of the code base .

AI usage policy:

You don’t need to use any of the AI tools, a pen and paper is all it needs.

AI usage is allowed for literature review, brain-storming, site building, and generating boiler-plates code. However, I am expecting personal validation and understanding of any AI generated code that is crucial to the project. For example, I won’t question your html skills of changing font or color, or your python skills of curve fitting and graphs making, but If there are codes involved in your project, you will be asked about the implementation of a physical engine/model in your site, or if you have tested the edge case, or how did you fine-tuned an existing model.